Contracts to Effects Compiler Correctness

(Local Handler)

Here, we provide the proof for a local handler version. Instead of leaving the handler outside, a flat contract is compiled into an effect and handle it locally. All other rules stays the same. Semantics of handler can be either Deep or Shallow. For this proof, we choose the Effects semantics to be Shallow Handler for simplicity. Deep Handler version has a pushes a handler each time a check is performed, since the installation of the handler is automatic. These handler will not be invoked, therefore it buries the current execution until it reaches a value and remove all of them at once. Because the effect is local, and activates once only, using Shallow Handler is optimal (while also easier to prove).

$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail^{l} e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail^{l} e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail^{l} \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, (\mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4\rangle\rangle))}\ \text{Compile-Flat}$$

$$\begin{aligned} \mathtt{h}_{\mathscr{C}} : \mathtt{lab} &= \{\mathscr{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\} \\ &= \{\mathscr{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if } e\ v \text{ then } k\ v \text{ else } \mathtt{error}_l\} \\ &\text{where } \lambda\langle l, \langle e, v\rangle\rangle \text{ deconstructs the argument as tuple pattern} \end{aligned}$$

Relations

$R_v \subseteq \mathsf{Contracts}\ v \times \mathtt{Effects}\ v$

Abuse of notation: If $(v, v') \in R_v$, we write $v \sim_{R_v} v'$ or $v \sim v'$. We write $\tilde{v}$ for any value $\tilde{v}$ satisfying $v \sim \tilde{v}$.

$$\begin{aligned} x &\sim x \\ b &\sim b \\ N &\sim N \\ \lambda x : \tau.e &\sim \lambda x : \tau^*.\tilde{e} \\ \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) &\sim \tilde{e} \end{aligned}$$

$$
\begin{array}{rcl}
 & & e = \lambda x : \tau.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) & \sim & \tilde{e} \\
 & & e = \lambda x : \tau.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
 & & \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]
\end{array}
$$

$\boxed{R_e \subseteq \mathsf{Contracts}\ e \times \mathtt{Effects}\ e}$

Abuse of notation: If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We write $\tilde{e}$ for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

$$
\begin{array}{rcl}
v & \sim & \tilde{v} \\
e_1\ e_2 & \sim & \tilde{e}_1\ \tilde{e}_2 \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) & \sim & \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, (\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle)) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
 & & \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right] \\
\mathsf{blame}_j^k & \sim & \mathtt{error}_k
\end{array}
$$

$\boxed{R_E \subseteq \mathsf{Contracts}\ E \times \mathtt{Effects}\ E}$

Abuse of notation: If $(E, E') \in R_E$, we write $E \sim_{R_E} E'$ or $E \sim E'$. We write $\tilde{E}$ for any expression $\tilde{E}$ satisfying $E \sim \tilde{E}$.

$$
\begin{array}{rcl}
\square & \sim & \square \\
E\ e & \sim & \tilde{E}\ \tilde{e} \\
v\ E & \sim & \tilde{v}\ \tilde{E} \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E) & \sim & \mathtt{handle}_s\left(\mathtt{h}_{\mathscr{C}}, \left(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{E} \rangle\rangle\right)\right) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E) & \sim & \tilde{e}_1\ \tilde{E} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E) & \sim & \tilde{e}_1\ \tilde{E} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)
\end{array}
$$

$$\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$$
$$E\left[\mathsf{check}_j^k(E', v)\right] \quad \sim \quad \text{if } \tilde{E}' \text{ then } \left(\lambda x.\tilde{E}[x]\right) \tilde{v} \text{ else } \mathtt{error}_k$$

Lemma 0.1. (Relation is closed)

If $e \neq \mathsf{check}_j^k(e_0, v_0)$ and $e \sim e'$ and $E \sim E'$ then $E[e] \sim E'[e']$

Proof.

By induction on the structure of $E \sim E'$.

- Case $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), E_1) \sim \mathtt{handle}_s\left(\mathtt{h}_{\mathscr{C}}, \left(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}_1\rangle\rangle\right)\right)$

  $E_1[e] \sim \tilde{E}_1[e']$ I.H.
  $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), E_1[e]) \sim \mathtt{handle}_s\left(\mathtt{h}_{\mathscr{C}}, \left(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}_1[e']\rangle\rangle\right)\right)$
  $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), E_1)[e] \sim \left(\mathtt{handle}_s\left(\mathtt{h}_{\mathscr{C}}, \left(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}_1\rangle\rangle\right)\right)\right)[e']$

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E_1) \sim \tilde{e}_1\ \tilde{E}_1$ and $e_1 = \left(\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)\right)$

  $E_1[e] \sim \tilde{E}_1[e']$ I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E_1[e]) \sim \tilde{e}_1\ \tilde{E}_1[e']$
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E_1)[e] \sim \left(\tilde{e}_1\ \tilde{E}_1\right)[e']$

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E_1) \sim \tilde{e}_1\ \tilde{E}_1$ and $e_1 = \left(\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$

  $E_1[e] \sim \tilde{E}_1[e']$ I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E_1[e]) \sim \tilde{e}_1\ \tilde{E}_1[e']$
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E_1)[e] \sim \left(\tilde{e}_1\ \tilde{E}_1\right)[e']$

- Case $E'\left[\mathsf{check}_l^k(E_1, v_1)\right] \sim \text{if } \tilde{E}_1 \text{ then } \left(\lambda x.\tilde{E}'[x]\right) \tilde{v}_1 \text{ else } \mathtt{error}_k$

  $E_1[e] \sim \tilde{E}_1[e']$ I.H.
  $E'\left[\mathsf{check}_l^k(E_1[e], v_1)\right] \sim \text{if } \tilde{E}_1[e'] \text{ then } \left(\lambda x.\tilde{E}'[x]\right) \tilde{v}_1 \text{ else } \mathtt{error}_k$
  $E'\left[\mathsf{check}_l^k(E_1, v_1)\right][e] \sim \left(\text{if } \tilde{E}_1 \text{ then } \left(\lambda x.\tilde{E}'[x]\right) \tilde{v}_1 \text{ else } \mathtt{error}_k\right)[e']$

□

Lemma 0.2. (Relation substitution)

If $e \sim e'$ and $v \sim v'$ then $e[x := v] \sim e'[x := v']$

Proof. □

Lemma 0.3. (Find Relation Context)

If $e_1 = E[e_2]$ and $e_2 \neq \mathsf{check}_j^k(e_{2a}, v)$ and $e_1 \sim e_1'$ then there exists $\tilde{E}, \tilde{e}_2$, such that $e_1' = \tilde{E}[\tilde{e}_2]$

Proof.

By induction on E

- Case $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E_1)$

  $e_1 = \mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E_1[e_2]) = \mathsf{mon}_j^{k,l}(\mathsf{flat}(e), e_3)$
  $e_1 \sim \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, (\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{e}_3 \rangle\rangle))$
  $\tilde{e}_3 = \tilde{E}_1[\tilde{e}_2]$ I.H. by $e_3 = E_1[e_2] \sim \tilde{e}_3$
  $e_1' = \mathtt{handle}_s\big(\mathtt{h}_{\mathscr{C}}, \big(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{E}_1 \rangle\rangle\big)\big)[\tilde{e}_2]$
  $\tilde{E} = \mathtt{handle}_s\big(\mathtt{h}_{\mathscr{C}}, \big(\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{E}_1 \rangle\rangle\big)\big)$ Choose

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E_1)$

  $e_1 = \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E_1[e_2]) = \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_3)$
  $e_1 \sim \big(\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big)\big)\ \tilde{e}_3$
  $\tilde{e}_3 = \tilde{E}_1[\tilde{e}_2]$ I.H. by $e_3 = E_1[e_2] \sim \tilde{e}_3$
  $e_1' = \big(\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big)\big)\ \tilde{E}_1[\tilde{e}_2]$
  $\tilde{e} = \big(\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big)\big)\ \tilde{E}_1$ Choose
  Where $f, y \neq x$ and $f, y \notin \mathrm{fv}(e_2)$ by alpha-renaming

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, E_1)$

  $e_1 = \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, E_1[e_2])$
  $= \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_3)$
  $e_1 \sim \big(\lambda f.\lambda y.\mathsf{mon}_j^{k,l}\big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\big)\big)\ \tilde{e}_3$
  $\kappa_3 = \kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\big]$
  $\tilde{e}_3 = \tilde{E}_1[\tilde{e}_2]$ I.H. by $e_3 = E_1[e_2] \sim \tilde{e}_3$
  $e_1' \sim \big(\lambda f.\lambda y.\mathsf{mon}_j^{k,l}\big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\big)\big)\ \tilde{E}_1[\tilde{e}_2]$
  $\tilde{E} = \big(\lambda f.\lambda y.\mathsf{mon}_j^{k,l}\big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\big)\big)\ \tilde{E}_1$ Choose
  Where $f, y \neq x$ and $f, y \notin \mathrm{fv}(e_2)$ by alpha-renaming

- Similarly for other cases.

□

Lemma 0.4. (Find Relation Context for check)

If $e_1 = E\big[\mathsf{check}_j^k(e_2, v)\big]$ and $e_1 \sim e_1'$ then there exists $\tilde{E}, \tilde{e}_2, \tilde{v}$, such that $e_1' =$ if $\tilde{e}_2$ then $\big(\lambda x.\tilde{E}[x]\big)\ \tilde{v}$ else $\mathtt{error}_k$

Proof.

By induction on the structure of E.

- Case $\square$

  $e_1 = \mathsf{check}_j^k(e_2, v)$

$e_2 = E_1[e_3]$

- Case $e_3 \neq \mathsf{check}^{k_0}_{j_0}(e_4, v_0)$

  $\tilde{e}_2 = \tilde{E}_1[\tilde{e}_3]$ Lemma 0.3

  $e_1 \sim \text{if } \tilde{e}_2 \text{ then } (\lambda x.x)\ \tilde{v} \text{ else } \mathtt{error}_k$

  $\tilde{E} = \square$ Choose

- Case $e_3 = \mathsf{check}^{k_0}_{j_0}(e_4, v_0)$

  $\tilde{e}_2 = \text{if } \tilde{e}_4 \text{ then } \left(\lambda x.\tilde{E}_1[x]\right)\ \tilde{v}_0 \text{ else } \mathtt{error}_k$ I.H.

  $e_1 \sim \text{if } \tilde{e}_2 \text{ then } (\lambda x.x)\ \tilde{v} \text{ else } \mathtt{error}_k$

  $\tilde{E} = \square$ Choose

- Case $\mathsf{mon}^{k_0,l_0}_{j_0}(\mathsf{flat}(e_0), E_0)$

  $e_1 = \mathsf{mon}^{k_0,l_0}_{j_0}\left(\mathsf{flat}(e_0), E_0\left[\mathsf{check}^{k}_{j}(e_2, v)\right]\right)$

  $= \mathsf{mon}^{k_0,l_0}_{j_0}(\mathsf{flat}(e_0), E_0)\left[\mathsf{check}^{k}_{j}(e_2, v)\right]$

  $= E_1\left[\mathsf{check}^{k}_{j}(e_2, v)\right]$

  $e_1 \sim \text{if } \tilde{e}_2 \text{ then } \left(\lambda x.\tilde{E}_1[x]\right)\ \tilde{v} \text{ else } \mathtt{error}_k$ I.H.

  $\tilde{E} = \tilde{E}_1$ Choose

□

Lemma 0.5. (Compiler is Relation)

If $\Gamma \vdash e : \tau \rightarrowtail^{l} e'$ then $e \sim e'$

Proof. □

Correctness Theorem

Theorem 0.6.

If $e \twoheadrightarrow^{*} b$ and $\cdot \vdash e : \tau \rightarrowtail^{l} e'$ then $e' \Longrightarrow^{*} b$

Proof.

$e \sim e'$ Lemma 0.5

$e' \Longrightarrow^{*} \tilde{b} = b$ Lemma 0.7

□

Lemma 0.7.

If $e \sim e'$ and $e \twoheadrightarrow^{*} r$ then $e' \Longrightarrow^{*} \tilde{r}$.

Where $r = v \mid \mathsf{blame}^{k}_{j}$ and $\tilde{r} = \tilde{v} \mid \mathtt{error}_k$

Proof.

By induction on the structure of $e \twoheadrightarrow^* r$

- Case $r \twoheadrightarrow^* r$

  $e \sim e'$ — Given
  $e' = \tilde{r}$ — Inversion
  $e' \Longrightarrow^* \tilde{r}$ — Reflexivity

- Case $e \twoheadrightarrow e_1$ and $e_1 \twoheadrightarrow^* r$

  $e' \Longrightarrow^* e'_1$ — Lemma 0.8
  $e_1 \sim e'_1$ — Lemma 0.8
  $e'_1 \Longrightarrow^* \tilde{r}$ — I.H.
  $e' \Longrightarrow^* \tilde{r}$ — Step

□

Lemma 0.8.

If $e_1 \sim e'_1$ and $e_1 \twoheadrightarrow e_2$ then $e'_1 \Longrightarrow^* e'_2$ and $e_2 \sim e'_2$

Proof.

By case analysis on $e_1 \twoheadrightarrow e_2$.

- $E\left[\mathsf{blame}_j^k\right] \twoheadrightarrow \mathsf{blame}_j^k$, and $E \neq \square$

  $E \sim \tilde{E}$ — Lemma 0.3
  $e'_1 = \tilde{E}[\mathtt{error}_k]$
  $e'_1 \Longrightarrow^* \mathtt{error}_k$ — Step and Reflexivity
  $e'_2 = \mathtt{error}_k$ — Choose

- $E[e_3] \twoheadrightarrow E[e_4]$ where $e_3 \dashrightarrow e_4$ is a redex

  - Case $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}_j^k(e\ v, v)$

    $E \sim \tilde{E}$ — Lemma 0.3
    $e_3 \sim \tilde{e}_3$ — Lemma 0.3
    $\tilde{e}_3 = \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, (\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v} \rangle\rangle))$ — Inversion
    $e'_1 = \tilde{E}[\tilde{e}_3]$
    $e'_1 \Longrightarrow \text{if } \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\tilde{E}[x]\right)\ \tilde{v} \text{ else } \mathtt{error}_k$
    $E[e_4] \sim \text{if } \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\tilde{E}[x]\right)\ \tilde{v} \text{ else } \mathtt{error}_k$
    $e'_2 = \text{if } \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\tilde{E}[x]\right)\ \tilde{v} \text{ else } \mathtt{error}_k$ — Choose

  - Case $\mathsf{mon}_j^{k,l}(\kappa, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa, v)$ and $\kappa \neq \mathsf{flat}(e)$

    $E \sim \tilde{E}$ — Lemma 0.3
    $e_3 \sim \tilde{e}_3$ — Lemma 0.3

- Case $\kappa = \kappa_1 \to \kappa_2$

  $\tilde{e}_3 = \tilde{e}_5\ \tilde{v}$ — Inversion
  $e_5 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) = \lambda f.e_{5a}$
  $e_1' = \tilde{E}[\tilde{e}_5\ \tilde{v}]$
  $e_1' \Longrightarrow \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$
  $e_{5a} \sim \tilde{e}_{5a}$
  $v \sim \tilde{v}$
  $e_{5a}[f := v] \sim \tilde{e}_{5a}[f := \tilde{v}]$ — Lemma 0.2
  $e_{5a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$
  $e_4 \sim \tilde{e}_{5a}[f := \tilde{v}]$ — Relation
  $e_2 = E[e_4] \sim \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$ — Lemma 0.1
  $e_2' = \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$ — Choose

- Case $\kappa = \kappa_1 \to_i \lambda y.\kappa_2$

  $\tilde{e}_3 = \tilde{e}_5\ \tilde{v}$ — Inversion
  $e_5 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) = \lambda f.e_{5a}$
  $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$
  $e_1' = \tilde{E}[\tilde{e}_5\ \tilde{v}]$
  $e_1' \Longrightarrow \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$
  $e_{5a} \sim \tilde{e}_{5a}$
  $v \sim \tilde{v}$
  $e_{5a}[f := v] \sim \tilde{e}_{5a}[f := \tilde{v}]$ — Lemma 0.2
  $e_{5a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$
  $e_4 \sim \tilde{e}_{5a}[f := \tilde{v}]$ — Relation
  $e_2 = E[e_4] \sim \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$ — Lemma 0.1
  $e_2' = \tilde{E}[\tilde{e}_{5a}[f := \tilde{v}]]$ — Choose

▸ Case $\mathsf{check}_j^k(\mathsf{true}, v) \dashrightarrow v$

$E[e_3] \sim \mathsf{if}\ \mathsf{true}\ \mathsf{then}\ \left(\lambda x.\tilde{E}[x]\right)\tilde{v}\ \mathsf{else}\ \mathtt{error}_k$ — Lemma 0.4
$e_1' \Longrightarrow^* \tilde{E}[\tilde{v}]$
$e_2 \sim \tilde{E}[\tilde{v}]$
$e_2' = \tilde{E}[\tilde{v}]$ — Choose

▸ Case $\mathsf{check}_j^k(\mathsf{false}, v) \dashrightarrow \mathsf{blame}_j^k$

$E[e_3] \sim \mathsf{if}\ \mathsf{false}\ \mathsf{then}\ \left(\lambda x.\tilde{E}[x]\right)\tilde{v}\ \mathsf{else}\ \mathtt{error}_k$ — Lemma 0.4
$e_1' \Longrightarrow^* \mathtt{error}_k$
$e_2 \sim \mathtt{error}_k$
$e_2' = \mathtt{error}_k$ — Choose

□